

CASEY BOARD INTELLIGENCE PACK

Space Base has been classified as Surface Habitat/Base in Moon / Lunar Surface. CASEY estimates \$198.0B P50 cost, \$158.4B to \$257.4B range, 246 month baseline, High risk and 65% confidence under the Lower Risk scenario.

Project	Space Base	Scenario	Lower Risk
P50	\$198.0B	Cost Range	\$158.4B - \$257.4B
QCRA P80	\$274.1B	QSRA P80	267.0 months
Direct / Indirect / Reserve	\$154.4B / \$29.7B / \$23.8B	Risk / Confidence	High / 65%

Board Recommendation

- Lower-risk selected: more assurance, surveys, float and contingency raise headline cost but improve confidence.
- Controls consequence: stronger evidence should move risk from unknown unknowns into quantified known unknowns.
- Board decision: approve with a confidence-improvement plan and gate evidence pack.

QCRA and QSRA - Separate Tornado Drivers

Rank	QCRA Cost Driver	CBS	Mean Cost	QSRA Schedule Driver	Activity	Mean Days
1	Life-support reliability	01.06	\$7.3B	Launch manifest delay	A1400	24.6
2	Mass growth	01.05	\$7.0B	Life-support reliability	A1600	23.5
3	Launch manifest delay	01.04	\$7.0B	Permits / approvals delay	A1900	17.2
4	Market escalation	01.08	\$5.4B	Mass growth	A1500	17.1
5	Scope growth	01.07	\$4.2B	Scope growth	A1700	14.2
6	Design maturity gap	02.01	\$4.1B	Productivity underperformance	P1000	13.2
7	Productivity underperformance	02.03	\$3.6B	Design maturity gap	A2000	12.9
8	Supply chain delay	02.02	\$3.4B	Supply chain delay	A2100	12.7
9	Interface misalignment	03.02	\$2.8B	Interface misalignment	P3000	9.3
10	Commissioning delay	03.01	\$2.3B	Commissioning delay	P2000	8.8

Risk Register - Board Top Risks

ID	Risk	Cause	Event	Impact	Owner	Response
R-S01	Launch manifest delay	Launch provider capacity, weather windows, manifest deadline which reduces payload mass	Confirmed launch which reduces mass	Programme delay, storage cost, resequencing and mission readiness risk	Launch Integration Lead	Reserve alternate launch slot, freeze payload interface, maintain manifest
R-S03	Life-support reliability	ECLSS reliability, spares or redundancy evidence	Life support system fails	Reduced test exposure, launch delay, redesign and mission failure	Life Support Lead	Prototype testing, redundancy, reliability growth, independent review
R-S02	Mass growth	Design creep, shielding, redundancy or late payload addition increases mass	Final addition increases mass and landing allowance	Launch cost, redesign, performance loss and mission safety risk	Chief Engineer	Mass control board, margin policy, design-to-mass reviews
R-002	Market escalation	Supplier market tightness, inflation, FX and long lead times	Prices increase as demand escalation profile	P50 cost becomes understated; commercial approvals or programme delay	Commercial Strategy Manager	Early procurement, index-linked allowances, market testing, alternative suppliers
R-004	Design maturity gap	Estimate is based on immature design, incomplete design	Design changes during detailed design	Cost growth, quantity movement, rework and schedule resourcing	Design Manager	Design maturity gates, independent review, assumption register
R-001	Scope growth	Ambiguous requirements, immature scope freeze	Approved scope expands during estimate freeze	Additional quantities, redesign, rework, procurement churn	Project Board	Early freeze scope baseline, change board, value gates, client delivery
R-006	Productivity underperformance	Access constraints, poor sequencing, labour scarcity	Activity producing rates underperform plan	Extended duration, increased preliminaries and loss of float	Delivery Lead	Package productivity controls, daily planning, earned value control
R-005	Supply chain delay	Long-lead equipment, constrained suppliers or supplier promise dates	Supplier promise dates move right	Delayed installation, critical path movement and acceleration cost	Procurement Lead	Alternate suppliers, early orders, expediting, framework options
R-008	Interface misalignment	Multiple contractors, unclear interface ownership	Package interfaces do not align during installation	Rework, claims, delay and fragmented accountability	Integration Manager	Interface control documents, weekly interface board, accountability
R-007	Commissioning delay	Incomplete readiness, software integration issues	Systems failed during testing repeated commissioning	Delayed handover, operational readiness slippage and liquidation risk	Operations Manager	Commissioning readiness plan, early test packs, digital system

Cost Estimate - Selected Class Only

CBS	Description	Type	P10	P50	P90	Basis
01.01	DDT&E Engineering	Direct	\$14.3B	\$17.8B	\$23.2B	Engineering design, mission architecture and qualification
01.02	Systems Engineering & Integration	Direct	\$9.5B	\$11.9B	\$15.4B	Requirements, ICDs, verification and validation
01.03	Structures / Habitat Shell	Direct	\$15.8B	\$19.8B	\$25.7B	Pressure vessels, shielding and primary structures
01.04	Power Generation & Distribution	Direct	\$12.7B	\$15.8B	\$20.6B	Solar/nuclear source, grid and controls
01.05	ECLSS / Life Support	Direct	\$14.3B	\$17.8B	\$23.2B	Air, water, waste, atmosphere and safety systems
01.06	Payload / Mission Equipment	Direct	\$11.1B	\$13.9B	\$18.0B	Scientific, medical, industrial or settlement payload
01.07	Robotics / Autonomous Construction	Direct	\$9.5B	\$11.9B	\$15.4B	Robotic construction, inspection and maintenance
01.08	Launch Services	Direct	\$22.2B	\$27.7B	\$36.0B	Launch vehicle, manifest, payload processing
01.09	Transport / Landing / Surface Logistics	Direct	\$14.3B	\$17.8B	\$23.2B	Transit, landing, mobility and logistics chain
02.01	Test & Qualification Facilities	Indirect	\$7.9B	\$9.9B	\$12.9B	Thermal-vac, vibration, radiation and integrated testing
02.02	Safety & Mission Assurance	Indirect	\$7.9B	\$9.9B	\$12.9B	Quality, safety, independent review and compliance
02.03	Programme Management / PMO	Indirect	\$7.9B	\$9.9B	\$12.9B	Controls, reporting, commercial and integration PMO
03.01	Risk Contingency / UFE	Reserve	\$11.1B	\$13.9B	\$18.0B	Known-unknown risk allowance linked to QCRA/QSRA
03.02	Management Reserve	Reserve	\$7.9B	\$9.9B	\$12.9B	Sponsor-held unknown-unknown reserve